

FORCES

Invisible forces are constantly at play in our day-to-day life, from the wind rustling the leaves of trees to the tension in the cables of a suspension bridge. A force is any push or pull. Forces can change an object's speed or direction of motion, or can change its shape. English scientist Isaac Newton figured out how forces affect motion over three hundred years ago (see pp.98–99). His principles are still applied in many fields of science, engineering, and in daily life today.

WHAT IS A FORCE?

A force can be a push or a pull. Although you can't see a force, you can often see what it does. A force can change the speed, direction, or shape of an object. Motion is caused by forces, but forces don't always make things move—balanced forces are essential for building stability.

Contact forces

When one object comes into contact with another and exerts a force, this is called a contact force. Either a push or a pull, this force changes the direction, speed, or shape of the object.



Changing direction

If a player bounces a ball against a wall during practice, the wall exerts a force on the ball that changes its direction.



Changing speed

When a player kicks, back-heels, or volleys a soccer ball, the force that is applied changes the ball's speed.



Changing shape

Kicking or stepping on the soccer ball applies a force that momentarily squashes it, changing the ball's shape.

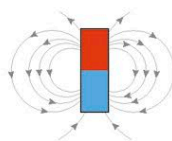
Non-contact forces

All forces are invisible, but some are exerted without physical contact between objects. The closer two objects are to each other, the stronger is the force.



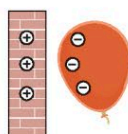
Gravity

Gravity is a force of attraction between objects with mass. Every object in the universe pulls on every other object.



Magnetism

A magnet creates a magnetic field around it. If a magnetic material is brought into the field, a force is exerted on it.



Static electricity

A charged object creates an electric field. If another charged object is moved into the field, a force acts on it.

Weight, gravity, and mass

Weight is not the same as mass, which is a measure of how much matter is in an object. Weight is the force acting on that matter and is the result of gravity. The mass of an object is the same everywhere, but its weight can change.



Measuring forces

Forces can be measured using a force meter, which contains a spring connected to a metal hook. The spring stretches when a force is applied to the hook. The bigger the force, the longer the spring stretches and the bigger the reading. The unit of force is the newton (N).



Calculating weight

Mass is measured in kilograms (kg). Weight can be calculated as mass $\times$ gravity (N/kg). The pull of gravity at Earth's surface is roughly 10 N/kg, so an object with a mass of 1 kg weighs 10N.

BALANCED AND UNBALANCED FORCES

Not all forces acting on an object make it move faster or in a different direction: forces on a bridge must be balanced for the structure to remain stable. In a tug of war, there's no winner while the forces are balanced; it takes a greater force from one team to win.

Balanced forces

If two forces acting on an object are equal in size but opposite in direction, they are balanced. An object that is not moving will stay still, and an object in motion will keep moving at the same speed in the same direction.



Unbalanced forces

If two forces acting on an object are not equal, they are unbalanced. An object that is not moving will start moving, and an object in motion will change speed or direction.

